

OCEAN HUNTERS ►

Shoaling fish are hunted by a variety of predatory fish such as these tuna, as well as other hunters such as dolphins. These are fast-moving animals that patrol large areas of ocean in search of prey. Tuna and dolphins hunt in packs, but others such as marlins and swordfish are solitary ocean nomads that stay on the move throughout their lives, searching for prey.



TOP PREDATORS ►

The hunters are hunted themselves by formidable predators like the great white shark. Such predators are relatively scarce, because it takes many tuna to feed one shark, and huge numbers of small fish to feed all the tuna. The tuna also use up a lot of energy as they swim, rather than turning it into flesh that the shark can eat. This means that each shark must be supported by many times its own weight in small fish.



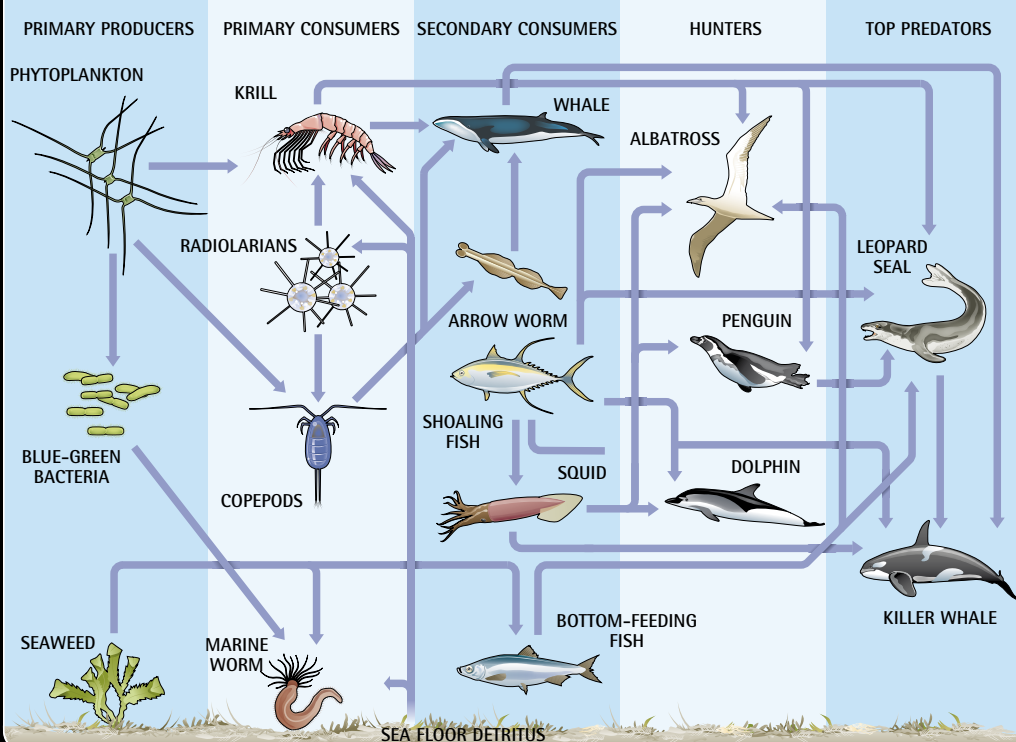
food web

GIANT FILTER-FEEDERS ▲

Some big fish eat very small animals, filter-feeding in the same way as shoaling fish. This is very efficient, because no energy is used up by intermediate feeders like tuna. It enables these animals to grow to immense sizes, and the filter-feeding whale shark seen above is the largest of all fish. Many whales feed in a similar way, and grow to a colossal size. The krill-eating blue whale, for example, is the biggest animal that has ever lived.



MARINE FOOD WEB (SOUTHERN OCEAN)



Every oceanic habitat has its own food web, which connects the primary food producers with the top predators. Basically, each animal occupies a different level in the ecosystem. Primary producers like phytoplankton are eaten by primary consumers such as copepods and krill. These are eaten by secondary consumers like squid and fish, which are hunted by animals such as penguins and dolphins. These may then fall prey to top predators like killer whales. Meanwhile, the remains of dead animals and plankton sink to the ocean floor, where detritus is recycled into essential nutrients by decomposers such as marine worms.

However, this oceanic food web for the Southern Ocean shows that the connections are more complicated. This is partly because most animals eat various kinds of foods. A killer whale, for example, will eat penguins when it gets the chance, but it also feeds on the squid and fish that penguins eat. It might also eat a leopard seal, which is another top predator that preys mainly on penguins. Giant filter-feeders like whales feed at the same level as tiny arrow worms, but they eat the arrow worms, too. Whales also have nothing to fear from hunters such as seabirds and penguins that occupy the next level.

Despite the complications, one thing is clear about all oceanic food webs. Every animal, big or small, depends on the food created by primary producers such as phytoplankton. If there are no primary producers there is no food, and no food web.